

Evaluation of GPS Precipitable Water over Canada and the IGS Network

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ABSTRACT

Precipitable water (PW) derived from the GPS zenith tropospheric delay (ZTD) is evaluated (as a first step toward variational data assimilation) through comparison with that of collocated radiosondes (RS_PW), operational analyses, and 6-h forecasts (from the Canadian Global Environmental Multiscale model) of the Canadian Meteorological Centre. Two sources of ZTD data are considered: 1) final ZTD (over Canada), computed by the Geodetic Survey Division (GSD) of Natural Resources Canada, and 2) final ZTD (distributed globally), obtained from the International GPS Service (IGS). The mean GSD GPS-derived PW (GPS_PW) is 14.9 mm (reflecting the relatively cold Canadian climate), whereas that of the IGS dataset is 20.8 mm. Intercomparison statistics [correlation, standard deviation (SD), and bias] between GPS_PW and RS_PW are, respectively, 0.97, 2.04 mm, and 1.35 mm for the GSD data and 0.98, 2.6 mm, and 0.67 mm for the IGS data. Comparisons of GPS_PW with 6-h forecast PW (TRIAL_PW) show slightly lower correlations and a higher SD. The increase in SD is greater for the IGS data, which is not surprising, because in regions such as the Tropics and subtropics, moisture forecasts are of a lower quality and the RS observation network is sparse. From a three-way intercomparison (IGS GPS_PW, RS_PW, and TRIAL_PW) of the SD statistics, it is found that GPS_PW has the lowest estimated PW error (≈ 1 mm) for PW in the 5–30-mm range. For PW greater than 30 mm, the RS_PW estimated error is ≈ 2 mm, and that of GPS_PW is ≈ 2.5 mm. The TRIAL_PW estimated error increases with PW, reaching 5.5 mm in the 40–55-mm PW range. These intercomparison results indicate that GPS_PW should be a useful source of humidity information for NWP applications.

1. Introduction

An accurate representation of water vapor in weather forecast models is essential. Until the advent of space-based remote sensing, the main source of upper-air atmospheric humidity observations was the radiosonde (RS). While RS observations have good vertical resolution, the sparse spatial distribution of RS stations and low reporting frequency limits their ability to resolve the horizontal structure of small-scale weather features. Space-based remote soundings (measuring the vertical distribution of water vapor) at IR and microwave wavelengths provide global coverage of vertical profiles of humidity, but with a low vertical resolution (e.g., Soden et al. 1994; Deblonde and English 2003). Moreover, due to the high and variable surface emis-

sivity over land, it is currently not possible to retrieve humidity in the lower troposphere. The IR remote sounding of water vapor can only be performed in clear skies. Precipitable water (PW) can be measured accurately over the oceans in nonprecipitating atmospheres (standard deviation of ≈ 2 mm with respect to RS observations) with passive microwave remote sensing techniques (Deblonde and Wagneur 1997; Deblonde 1999) because the surface emissivity of the ocean surface (≈ 0.6) is much lower than that over land (≈ 0.9).

Bevis et al. (1992) introduced a new approach for the remote sensing of PW. The method relies on the measurement of the atmospheric delay of radio signals (L band at frequencies of 1.22760 and 1.57542 GHz) that are transmitted from GPS satellites. As the radio waves travel from a GPS satellite transmitter to a ground-based GPS receiver, they are refracted by the ionosphere and the (electrically) neutral atmosphere. The ionospheric delay is frequency dependent and can be removed in the processing of data from dual-frequency GPS receivers. The remaining delay is referred to as the tropospheric delay and can be separated into two

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parts—the hydrostatic delay, which depends mainly on the surface pressure at the GPS receiver, and the wet delay, which depends strongly on the total amount of water vapor along the wave trajectory and weakly on the atmospheric temperature (e.g., Davis et al. 1985). There are between 6 and 12 GPS satellites visible at any given time and location on the earth. Their signals follow slant paths depending on the azimuth and elevation of each satellite. The zenith tropospheric delay (ZTD) at a GPS site is estimated over a predetermined time interval (e.g., 1 or 2 h) by mapping slant delays to zenith-equivalent values with a zenith angle-dependent mapping function, which is usually assumed to be azimuthally symmetric. At GPS sites where observations of surface pressure and temperature are also available, PW (hereinafter referred to as GPS_PW) can readily be derived from ZTD (Bevis et al. 1992, 1994). GPS_PW is available with a high temporal resolution (essentially continuous), and is not adversely affected by the presence of clouds or precipitation.

ZTD is assimilated into data assimilation (DA) systems either indirectly, by assimilating GPS_PW, or directly, by assimilating ZTD itself. Since 1998, GPS_PW is assimilated at the National Oceanic and Atmosphere Administration (NOAA) Forecast Systems Laboratory (FSL) in the rapid update cycle (RUC) regional analysis-forecast system over North America (Benjamin et al. 2004; Gutman et al. 2003b, 2004). Falvey and Beavan (2002) have assimilated GPS_PW with a mesoscale model centered over New Zealand. Examples of assimilation of ZTD measurements with a regional three-dimensional variational data assimilation (3DVAR) system can be found in Vedel and Huang (2004) and Higgins (2001). Simulated ZTD measurements have been assimilated in a four-dimensional variational data assimilation (4DVAR) system (De Pondeca and Zou 2001). The advantage of assimilating ZTD over GPS_PW is that collocated measurements of surface observations (such as surface pressure and temperature) at the GPS receiver are not required. In most countries, with the United States being an exception, these surface data are often not available. Impact studies performed by Gutman et al. (2003b, 2004), in which GPS_PW is assimilated, and Vedel and Huang (2004), in which ZTD is assimilated, show improved short-range precipitation forecasts, but with a short-lasting impact [<18 h in Vedel and Huang (2004), and <6 h in Gutman et al. (2003b)]. At the Canadian Meteorological Centre (CMC), it is planned to assimilate ZTD in regional 3DVAR and 4DVAR systems when the latter becomes operational.

To evaluate GPS_PW retrievals from current GPS measurement networks, with the goal to potentially increase these networks, researchers in several countries have performed intercomparison studies between different observation sources of PW and sometimes also NWP products (e.g., Köpken 2001; Bokoye et al. 2002;

Ohtani and Naito 2000; Feng et al. 2001; Cucurull et al. 2000; Haase et al. 2003; Baltink et al. 2002; Coster et al. 1996; Basili et al. 2001; Wolfe and Gutman 2000; Yang et al. 1999). These studies have shown that the SD of GPS_PW minus the PW derived from collocated radiosondes (RS_PW) ranges between 1.4 and 2.6 mm. In this study, an evaluation is performed of GPS_PW through comparisons with RS_PW, analysis PW (ANAL_PW), and 6-h forecast PW (TRIAL_PW). ANAL_PW and TRIAL_PW are obtained with the Canadian global 3DVAR weather analysis and forecast system. The intercomparison is based on two ZTD datasets. For the first dataset, ZTD was provided by the Geodetic Survey Division (GSD) of Natural Resources Canada (NRCan) for 22 Canadian GPS sites for the months of April, July, and October 2001, and January 2002. For 98% of the observations, GPS_PW is less than 35 mm, reflecting the relatively cold Canadian climate. To extend the study to higher values of PW, ZTD was obtained from the International GPS Service (IGS) tracking network of globally distributed GPS sites and, furthermore, 3 months of data (June, August, and September 2001) were added to the original 4 months.

The derivation of PW from GPS ZTD and a description of the ZTD data are provided, respectively, in sections 2 and 3. Section 4 details how RS_PW is computed from radiosondes collocated with GPS sites. Section 5 summarizes the computation of ANAL_PW and TRIAL_PW, while section 6 describes how surface pressure, temperature, and vapor-weighted mean temperature, needed to compute GPS_PW from ZTD, are obtained at the GPS sites. Section 7 presents the results of the PW intercomparison study. Conclusions are presented in section 8.

2. Derivation of precipitable water from ZTD

The derivation of PW from ZTD follows Bevis et al. (1992). ZTD at sea level ranges between 2.2 and 2.7 m and can be written as the sum of the zenith hydrostatic delay (ZHD) and the zenith wet delay (ZWD). About 90% of ZTD is due to ZHD, or the atmospheric mass component, which is directly related to the surface air pressure P_s at the GPS site. The commonly used equation for ZHD (mm) (Saastamoinen 1972) is

$$\text{ZHD}(\lambda, H, P_s) = 2.2779(1 - 0.00266 \cos 2\lambda - 0.00028H)^{-1} P_s, \quad (1)$$

where P_s is in hectopascals, λ is the latitude, and H is the height of the GPS site above the geoid in kilometers. ZWD is a function of the column-integrated water vapor, which is equal to PW divided by the density of water, and the vapor-weighted column mean temperature T_m above the GPS receiver. GPS_PW can be written as

$$\text{GPS_PW (mm)} = \frac{1}{\rho_w} \left[\frac{10^6}{(k_3 R_v / T_m) + k_2' R_v} \right] \times [\text{ZTD} - \text{ZHD}(\lambda, H, P_s)], \quad (2)$$

where k_3 and k_2' are constants (Bevis et al. 1994), and R_v is the gas constant of water vapor; T_m is defined as

$$T_m = \frac{\int_0^{Z_{\text{top}}} (e/T) dz}{\int_0^{Z_{\text{top}}} (e/T^2) dz}, \quad (3)$$

where e is the water vapor partial pressure, T is temperature, z is height (AGL), and Z_{top} is the height of the top of the atmosphere. Section 6 describes how T_m is computed for this study.

3. Zenith tropospheric delay data

a. GSD dataset

The GSD dataset consists of hourly estimates of ZTD for 22 GPS receivers located in Canada for the months of April, July, and October 2001, and January 2002 (to sample the four seasons). ZTD is computed with Bernese V4.2 software (Hugentobler et al. 2001) and the final IGS orbits, which are the most accurate but have a latency of about 2 weeks. ZTD is estimated after applying the dry Niell mapping function (Niell 1996) and assuming azimuthal symmetry. A satellite elevation cutoff angle of 5° is applied with an elevation-dependent weighting function of $1/\cos(z)^2$, where z is the satellite zenith angle. A problem with data from satellites with low elevation angles is that ground effects, such as GPS signal multipathing, may occur, where the signal is reflected from topographic features or other objects before reaching the receiver antenna. This is one of the reasons why low elevation satellites are down weighted.

ZTD (and GPS_PW derived from it) is not a point measurement, but rather is an estimated parameter from the least squares adjustment of the GPS observations valid over the area covered by the slant paths for a predetermined time period. At a height of 5 km, below which more than 90% of PW is found, a 5° (25°) satellite elevation cutoff angle results in averaging GPS observations over a cone with a base radius of ≈ 57 km (11 km). As is the case at any time, most of the satellites are above 25° , and the average volume is, thus, a cone with a base radius of ≈ 11 km.

Clock errors are eliminated by double differencing in the processing of the GPS data. A ZTD standard error (SE) was also provided with the GSD data, obtained as part of the least squares parameter estimation process. In this study, ZTD is rejected when it is outside the range of 1.4 and 2.8 m, or when its SE is greater than 10 mm (or the 4σ level).

b. IGS dataset

The IGS dataset consists of the IGS final combined tropospheric product (Gendt 1998). This product was chosen because it is available freely from IGS (http://www.gfz-potsdam.de/pb1/igs_trop_wg/index_IGS_TROP_WG.html), and it is also the most accurate. IGS provides 2-hourly estimates of ZTD for a subset of the IGS tracking network sites. ZTD is obtained by combining the ZTD solutions provided by each of seven global analysis centers (ACs) and is referred to here as the AC ZTD. Each AC (listed in Table 1) estimates ZTD for its own set of IGS stations, with its own strategy, and with its own final orbits. Therefore, not all ACs contribute ZTD solutions for all sites or times. The AC ZTD is accompanied by the standard deviation (AC ZTD SD) of the ensemble of ZTD solutions, which serves as an indicator of overall agreement. The AC ZTD product states an accuracy of ≈ 4 mm, which is the long-term AC ZTD SD. This corresponds to roughly 0.6 mm in PW [see Table 2, which summarizes the results of an error analysis of GPS_PW based on Eq. (2)]. In this study, AC ZTD data with an AC ZTD SD greater than 15 mm (or the 4σ level) are rejected.

Over several years AC solutions (including the 7-month period studied here) reveal that ZTD solutions for two of the seven ACs [European Space Agency (ESA) and National Geodetic Survey (NGS)] are significantly different from the others (G. Gendt 2003, personal communication). Therefore, in this study, AC ZTD data are rejected when ESA and/or NGS are the sole contributors. For the 7 months of data considered, AC ZTD is available for 186 of the 368 IGS global tracking network sites.

4. Radiosonde precipitable water

By integrating specific humidity q with respect to pressure, RS_PW is computed from radiosonde observations (reported 2 times per day at 0000 and 1200 UTC). The RS quality control (QC) is discussed in ap-

TABLE 1. The seven analysis centers of IGS that contribute solutions to the IGS final ZTD product.

Organization	ID
Center for Orbit Determination in Europe, Switzerland	COD
European Space Operations Center, ESA (European Space Agency), Germany	ESA
GeoForschungsZentrum, Germany	GFZ
Jet Propulsion Laboratory, United States	JPL
National Oceanic and Atmospheric Administration/National Geodetic Survey (NGS), United States	NGS
Natural Resources Canada/GSD, Canada	EMR*
Scripps Institution of Oceanography, United States	SIO

* The previous name of Natural Resources Canada was Energy Mines and Resources.

TABLE 2. Examples of GPS_PW error [computed with Eq. (2)] resulting from given errors in ZTD, P_s , and T_m . Absolute and relative errors (in percent) are for a dry (PW of 4 mm) and moist (PW of 40 mm) atmosphere.

Variable	Error	Error in PW computed with Eq. (2)	
		Dry atmosphere	Moist atmosphere
ZTD	4 mm (final ZTD product)	0.6 mm or 15%	0.6 mm or 2%
P_s	1 hPa* ($P_s = 1000$ hPa)	0.4 mm or 10%	0.4 mm or 1%
T_m	5 K or 1.8% ($T_m = 273$ K)	0.07 mm or 1.8%	0.72 mm or 1.8%
Sum of errors		1.07 mm or 27%	1.72 mm or 4%

* Surface pressure observations errors are discussed in appendix A.

pendix B. A GPS site and RS station are determined to be collocated if the horizontal distance between the two is less than 100 km. For RS stations located below the GPS sites (i.e., $DZ > 0$, where DZ is the GPS site height minus the RS station height), the vertical integration of RS q is started at a pressure level corresponding to the GPS site elevation. The pressure at this level is computed from the heights of the GPS and RS sites, and the hydrostatic approximation. At the starting pressure level, q is obtained by linear interpolation of the RS data. For RS stations located above GPS sites (i.e., $DZ < 0$), RS surface data are extrapolated to the level of the GPS site. Pressure is extrapolated hydrostatically with an assumed temperature lapse rate of $-6.5^\circ\text{C km}^{-1}$, while q is extrapolated by assuming that the dewpoint temperature depression (temperature minus dewpoint temperature) is constant with height.

To minimize interpolation/extrapolation errors, RS_PW at IGS sites is only computed when DZ is between -50 and 500 m. At the GSD sites, the DZ thresholds are -350 and 600 m. Because of the low number of observations available in the GSD dataset, a wider

range of acceptable DZ is applied to retain a sufficient number of GPS and RS data collocations. In Table 3 RS DZ is given for the GSD sites, and in Table 4 RS DZ is given for the IGS sites.

Vertical profiles of q obtained from the RS data are not exactly vertical, because the balloon drifts with the prevailing winds as it rises. For average conditions, the radiosonde has drifted 5 km horizontally by the time it reaches an altitude of 5 km (Wolfe and Gutman 2000).

5. Analysis and trial fields

In the CMC 3DVAR (Gauthier et al. 1999), observations for a given time are combined in an optimal way with a 6-h forecast valid at the same time. The 6-h forecast is produced with the Global Environmental Multiscale (GEM) model (Côté et al. 1998), and is referred to as the trial field (TRIAL). The final product obtained from the combination of observations and forecast is called the analysis (ANAL), which in turn provides the initial conditions for the next 6-h forecast,

TABLE 3. Site information for the 20 GSD/NRCan GPS sites. Surface pressure data sources (section 6) are GPS (GPS Met data), SYNO (surface synoptic reports), and TRIAL (TRIAL surface data). TRIAL DZ is GPS site minus TRIAL field elevation, RS DZ is GPS site minus RS elevation, and RS DX is the horizontal separation between GPS site and RS.

Name	Location	Elev MSL (m)	Surface pressure data source	TRIAL DZ (m)	RS DZ (m)	RS DX (km)	Collocated RS synoptic No.
CHUR	Churchill, Manitoba	29	GPS	2.5	0.4	3.3	71913
STJO	St. John's, Newfoundland	144	GPS	94.1	4.4	5.3	71801
CAGS	Gatineau, Quebec	269	TRIAL	92.1	76.7	80.8	71722
NRC1	Ottawa, Ontario	116	GPS	-49.1	-76.1	98.4	71722
ELIZ	Eliza Dome, British Columbia	182	TRIAL	-28.9	159.3	91.5	71109
BCOV	Beaver Cove, British Columbia	53	SYNO	-288.6	29.6	40.8	71109
HOLB	Holberg, British Columbia	576	TRIAL	521.9	553.2	53.4	71109
ALBH	Victoria, British Columbia	51	GPS	-113.2	-8.0	92.7	72797
ALGO	Algonquin Park, Ontario	238	GPS	-60.7			
PRDS	Calgary, Alberta	1264	GPS	-208.5			
SCH2	Schefferville, Quebec	518	GPS	-69.8			
WINN	Winnipeg, Manitoba	248	GPS	-1.0			
YELL	Yellowknife, Northwest Territories	8	GPS	27.4			
DUBO	Lac Du Bonnet, Manitoba	275	SYNO	10.5			
NANA	Nanosee Bay, British Columbia	24	SYNO	-297.0			
NTKA	Nootka Island, British Columbia	17	SYNO	-332.0			
FLIN	Flin Flon, Manitoba	351	TRIAL	27.2			
UCLU	Ucluelet, British Columbia	29	TRIAL	-172.3			
WHIT	Whitehorse, Yukon	1420	TRIAL	311.7			
WILL	Williams Lake, British Columbia	1110	TRIAL	69.9			

and so on, to produce what is called an analysis cycle. Both ANAL and TRIAL contain P_s and 3D fields of temperature and specific humidity at the GEM model grid points. The GEM model has a terrain-following coordinate system, a uniform latitude–longitude grid (0.9° or ≈ 100 km at the equator), and 28 vertical levels from the surface to the 10-hPa level.

The GEM gridded surface topography is obtained by averaging real surface terrain features over the model grid. Thus, at any given location, there can be significant differences between the real elevation and that of the GEM model. The differences between the GPS site elevations and those of the GEM model (TRIAL DZ) are given in Tables 3 and 4 for the GSD and IGS sites, respectively.

The P_s , q , and T from the ANAL and TRIAL fields are interpolated to the GPS site locations. ANAL_PW and TRIAL_PW are computed as described in section 4 for RS data with the same vertical interpolation/extrapolation method and DZ limits. TRIAL P_s at the GPS site elevation is also obtained with this method but is computed only at GPS sites where absolute TRIAL DZ is less than 800 m. The same limit of 800 m is applied operationally at CMC when verifying P_s observations against GEM model pressure.

6. Meteorological data at GPS sites (P_s , T_s , and T_m)

To compute PW from ZTD [Eq. (2)], P_s and T_m are needed. At some GPS sites, P_s , surface temperature T_s , and surface RH (referred to as GPS Met data) are reported by dedicated weather stations that are collocated with the GPS receivers. GSD and IGS GPS Met data are provided in the Receiver Independent Exchange (RINEX) format. GSD GPS Met data are reported at 15-min intervals, whereas the time interval for IGS GPS Met data is site dependent (up to 1 h). For the GSD (IGS) dataset, GPS Met data are available at 11 (46) of the 22 (186) sites.

GPS Met P_s is adjusted hydrostatically to the height of the GPS antenna when the height difference is known. For GPS sites without GPS Met observations, P_s (adjusted hydrostatically to GPS site elevation) is taken from a nearby surface synoptic weather station (when available), provided that it is located within 50 km of the GPS site and that the elevation difference between the two is less than 100 m. Gutman et al. (2003a) showed that when these conditions are relaxed, the accuracy of remote P_s becomes unsuitable for PW determination. The data reported at these stations are called SYNO data. SYNO P_s is available every 3 or 6 h. In appendix A, QC and instrument accuracy of GPS Met and SYNO P_s and comparisons with their ANAL and TRIAL field equivalents are discussed. When neither GPS Met nor SYNO P_s is available, TRIAL P_s is used (section 5). In this study, TRIAL P_s is chosen over

P_s from the analysis (ANAL P_s), because in a DA context, ANAL P_s will not be available.

After applying QC and the data-reduction procedures described so far, 20 sites (10 of which also have GPS Met data) remain for the GSD dataset and 112 GPS sites (31 of which also have GPS Met data) remain for the IGS dataset for which GPS_PW can be compared to RS_PW and/or TRIAL_PW (Tables 3 and 4).

Because T_m is a vapor-weighted mean temperature, and vapor content is concentrated in the lower atmosphere, T_m will be weighted to low-level temperature, which is generally well correlated with T_s . Linear regression equations relating T_m and T_s were derived from an ensemble of RS observations (e.g., Bevis et al. 1992; Bokoye et al. 2002). An rms of ≈ 5 K between T_m and T_s from RS observations was obtained by Bevis et al. (1992), and T_m can be predicted (with a regression equation) from T_s with an rms relative error less than 2% (Bevis et al. 1994). The sensitivity of PW to uncertainties in T_m is listed in Table 2. This sensitivity is small for dry atmospheres. For example, a 5-K temperature error at a T_m of 273 K and a PW of 4 mm yields a PW error of 0.07 mm. For low-moisture conditions, such as those commonly observed in Canada, the accuracy of GPS_PW depends more on the accuracy of the ZTD and P_s measurements than on that of T_m .

A more accurate estimate of T_m can be obtained from NWP forecasts (rms relative error less than 1% between T_m forecasts and collocated RS; Bevis et al. 1994). In this study, TRIAL data provide T_m for the calculation of GPS_PW, except for the results presented in section 7c, where GPS Met T_s and the regression equation of Bevis et al. (1992) yield T_m . Quality control on GPS Met T_s is performed by rejecting differences with T_s from the TRIAL (TRIAL T_s) that are larger than 2.5σ , where σ is the SD of GPS Met T_s minus TRIAL T_s computed over a month and at each site. TRIAL T_s is adjusted to the GPS site elevation by linear interpolation/extrapolation of TRIAL T using the same temperature lapse rate as in section 4.

7. Results of the precipitable water intercomparison

In sections 2–6, several factors were identified that are expected to affect the level of agreement between PW derived from different data sources (i.e., GPS_PW, RS_PW, ANAL_PW, and TRIAL_PW). In summary, these are errors in the observations themselves; differences in the volume of the atmosphere that is sampled (at a height of 5 km, a radius of ≈ 5 , ≈ 11 , and ≈ 45 km is sampled, respectively, for RS_PW, GPS_PW, and TRIAL_PW); and differences in observation location, time (at most ± 1 h difference for all data types), and surface elevation, and the associated adjustments and assumptions required to obtain PW at a common level

TABLE 4. Site information for the 112 IGS GPS sites. Surface pressure data sources (section 6) are GPS (GPS Met data), SYNO, and TRIAL. TRIAL DZ is the GPS site minus TRIAL elevation, RS DZ is the GPS site minus RS elevation, and RS DX is the horizontal separation between GPS site and RS. Missing GPS Met data: P = GPS Met data available but not used (either of insufficient quality or missing for ≥ 2 months); x = number of months (≤ 2) with missing data.

Name	Location	Elev MSL (m)	Surface pressure data source	TRIAL DZ (m)	RS DZ (m)	RS DX (km)	Three- way study sites	No. of months with missing ZTD	Missing GPS Met data
ALBH	Victoria, British Columbia, Canada	50.1	SYNO	-113.2	-8.0	92.7			P
AMC2	Colorado Springs, Colorado	1929.7	TRIAL	63.7	—	—			
AMMN	Amman, Jordan	1034.9	TRIAL	476.7	—	—		1	
ANKR	Ankara, Turkey	937.9	SYNO	-145.3	46.9	12.5			
AOML	Miami, Florida	27.9	GPS	27.9	22.9	21.9	X		
ARTU	Arti, Russia	254.3	TRIAL	-46.6	—	—			
ASC1	Ascension Island	91.6	TRIAL	91.5	—	—		6	
ATWC	Tsuna Alaska	85.0	TRIAL	-738.4	45.0	67.1		6	
AUCK	Whangaparaoa, New Zealand	97.4	GPS	27.7	70.4	27.4	X		1
BAHR	Manama, Bahrain	10.6	GPS	9.3	-1.4	82.9	X		
BARH	Bar Harbor, Maine	32.3	TRIAL	-24.9	—	—		3	P
BJFS	Beijing, China	98.4	GPS	-304.4	43.4	48.7			
BOR1	Borowiec, Poland	87.6	SYNO	-0.1	—	—			
BRMU	Bermuda, United Kingdom	23.8	SYNO	23.8	—	—			
BRUS	Brussels, Belgium	104.2	GPS	22.4	—	—			
BSHM	Binyamin_shm, Israel	204.6	TRIAL	56.1	169.6	88.6		1	
BUUC	Bucuresti, Romania	108.7	TRIAL	-55.36	17.7	45.0		1	
CAGL	Cagliari, Italy	191.9	TRIAL	63.5	—	—			
CAS1	Casey, Antarctica	38.6	SYNO	-172.0	-3.4	0.4			
CEDU	Ceduna, Australia	154.0	TRIAL	5.2	—	—			
CHAT	Waitangi, New Zealand	47.9	GPS	41.5	—	—			1
CHUR	Churchill, Manitoba, Canada	28.9	GPS	2.5	0.4	3.3	X		
CLGO	College, Alaska	184.7	SYNO	-238.8	46.7	6.0		6	
COCO	Cocos (Keeling), Australia	4.6	SYNO	4.6	0.6	1.0	X		
CRO1	Christiansted, U.S.A. Virgin Islands	13.5	TRIAL	4.7	—	—			
DAEJ	Taejon, Korea	92.0	TRIAL	-249.0	40.0	83.7			
DARW	Darwin, Australia	75.0	TRIAL	19.6	44.0	54.5			
DAV1	Davis, Antarctica	26.8	SYNO	-151.3	13.8	0.3			
DGAR	Diego Garcia Island, United Kingdom	9.7	TRIAL	9.7	—	—		5	
DRAG	Metzoki Dragot, Israel	100.9	TRIAL	-244.8	65.9	70.5			
DUBO	Lac du Bonnet, Manitoba, Canada	275.0	SYNO	10.5	—	—			
EISL	Easter Island, Chile	119.6	TRIAL	110.1	—	—			
ELRO	Kibutz El-ro, Israel	1058.8	TRIAL	427.2	—	—		1	
FAIR	Fairbanks, Alaska	307.5	GPS	-179.7	169.5	24.8			
FLIN	Flin Flon, Manitoba, Canada	351.0	TRIAL	27.1	—	—			
FORT	Eusebio (Fortaleza), Brazil	29.0	TRIAL	-21.7	—	—			
GILB	Gilb, Israel	485.2	TRIAL	158.6	450.2	77.3		1	
GLSV	Kiev, Ukraine	201.4	TRIAL	55.6	34.4	6.5			
GODE	Greenbelt, Maryland	48.0	SYNO	-6.6	—	—		1	
GOLD	Goldstone, California	1017.8	TRIAL	144.3	—	—			
GOPE	Ondrejov, Czech Republic	547.4	TRIAL	121.3	243.4	25.8			
GOUG	Gough Island, United Kingdom	57.9	SYNO	43.7	—	—		2	
GUAM	Dededo, Guam	147.1	SYNO	132.6	72.1	14.2	X		
GUAT	Guatemala City, Guatemala	1549.5	TRIAL	422.1	—	—			
HARB	Pretoria, South Africa	1529.8	TRIAL	205.7	—	—			
HERS	Hailsham, United Kingdom	31.5	GPS	-8.8	-20.5	3.8	X	2	2
HNLC	Honolulu, Hawaii	6.7	TRIAL	1.8	—	—		4	
HRAO	Krugersdorp, South Africa	1388.9	TRIAL	63.9	—	—			P
JOZE	Jozefoslaw, Poland	109.3	SYNO	-17.6	13.3	35.1	X		
KABR	Kabri, Israel	82.1	SYNO	-150.5	63.1	93.9			
KARR	Karratha, Australia	116.8	TRIAL	-13.7	—	—			
KERG	Port aux Français, Kerguelen Island	34.0	SYNO	-22.8	—	—			
KODK	Kodiak, Alaska	65.6	SYNO	-91.0	61.6	2.2			
KOSG	Kootwijk, Netherlands	54.5	TRIAL	47.9	39.5	43.8			
KSTU	Krasnoyarsk, Russia	248.7	TRIAL	-154.7	42.7	23.6		1	
KUNM	Kunming, China	2055.9	TRIAL	-2.0	163.9	11.8			
KWJ1	Kwajalein Atoll, Marshall Islands	9.0	SYNO	9.0	1.0	0.9	X		
LAMA	Olsztyn, Poland	157.7	SYNO	46.7	—	—			
MAC1	MacQuarie Island, Sub-Antarctic	14.1	SYNO	9.6	6.1	0.9	X		
MADR	Robledo, Spain	776.7	TRIAL	-61.7	143.7	57.2			

TABLE 4. (Continued)

Name	Location	Elev MSL (m)	Surface pressure data source	TRIAL DZ (m)	RS DZ (m)	RS DX (km)	Three- way study sites	No. of months with missing ZTD	Missing GPS Met data
MAG0	Magadan, Russia	344.7	TRIAL	104.6	228.7	2.9			P
MAS1	Maspalomas, Spain	155.6	TRIAL	67.3	119.6	97.3			
MATE	Matera, Italy	490.0	GPS	233.6	—	—			
MAW1	Mawson, Antarctica	29.9	SYNO	−492.6	13.9	0.5			
MCM4	Ross Island, Antarctica	151.3	TRIAL	125.3	—	—			
MDVO	Mendeleevo (Moscow), Russia	204.7	TRIAL	37.7	4.7	54.8			
METS	Kirkkonummi, Finland	76.2	GPS	56.2	−26.8	93.6	X		
NLIB	North Liberty, Iowa	239.8	TRIAL	3.0	10.8	84.2			
NOT1	Noto, Italy	85.1	TRIAL	−25.4	—	—			
NOUM	Noumea, France	23.4	TRIAL	−168.4	−48.6	4.1			
NPLD	Teddington, United Kingdom	26.6	TRIAL	−53.9	−25.4	73.9			
NRC1	Ottawa, Ontario, Canada	114.0	GPS	−49.0	—	—			
NRC2	Ottawa, Ontario, Canada	114.0	SYNO	−49.0	—	—			
NTUS	Singapore	71.9	TRIAL	51.7	55.9	33.4		1	
NVSK	Novosibirsk, Russia	160.6	TRIAL	−14.8	—	—			
OBER	Oberpfaffenhofen, Germany	605.3	TRIAL	35.0	116.3	24.9		5	
ONSA	Onsala, Sweden	10.1	TRIAL	−25.6	—	—			
PENC	Penc, Hungary	248.1	TRIAL	−25.2	108.1	40.7			
PERT	Perth, Australia	45.4	SYNO	−107.1	25.4	16.3		1	
PETP	Petropavlovsk-Kamchatka, Russia	188.5	TRIAL	−258.8	104.5	2.3			P
PIE1	Pie Town, New Mexico	2369.6	TRIAL	135.4	—	—			
POTS	Potsdam, Germany	133.7	SYNO	82.7	29.7	73.5	X		P
REYK	Reykjavik, Iceland	26.7	TRIAL	−119.7	−27.3	36.5			P
SELE	Almaty, Kazakhstan	1383.1	TRIAL	5.7	—	—			
SFER	San Fernando (Cadiz), Spain	41.2	SYNO	−0.6	37.2	84.2	X		
SHAO	Sheshan, China	11.9	SYNO	−14.1	7.9	42.1	X	1	
SJDV	Unknown (near Lyon), France	382.4	TRIAL	−18.0	142.4	35.4			
SSIA	San Salvador, El Salvador	663.2	TRIAL	208.0	—	—			
STJO	St. John's, Newfoundland, Canada	143.5	GPS	94.1	4.3	5.3	X		
SUWN	Suwon-shi, Korea	60.7	SYNO	−118.7	8.7	19.6			
SYOG	East Ongle Island, Antarctica	28.0	SYNO	−69.5	7.0	0.8			
THTI	Papeete, Tahiti	91.7	SYNO	−36.7	89.7	3.3	X		
THU1	Thule, Greenland	38.2	TRIAL	−247.6	−38.8	1.3		3	
TIXI	Tixi, Russia	53.7	TRIAL	−47.9	—	—			
TSKB	Tsukuba, Japan	28.4	TRIAL	−112.4	−2.6	7.3			
USNO	Washington, D.C.	82.0	SYNO	18.8	−16.0	33.4	X		P
USUD	Usuda, Japan	1466.0	TRIAL	337.5	—	—			
VILL	Villafranca, Spain	595.4	SYNO	−235.8	−37.6	32.1			
WES2	Westford, Massachusetts	114.5	GPS	−17.4	—	—			1
WHIT	Whitehorse, Yukon, Canada	1419.3	TRIAL	311.6	—	—			P
WILL	Williams Lake, British Columbia, Canada	1110.0	TRIAL	69.9	—	—			
WSRT	Westerbork, Netherlands	44.5	TRIAL	41.9	39.5	66.4			
WTZR	Koetzting, Germany	619.1	GPS	97.3	201.1	77.7	X		
WUHN	Wuhan City, China	107.8	SYNO	39.5	84.8	29.6	X		P
YAKT	Yakutsk, Russia	108.6	TRIAL	−49.8	7.6	2.4			P
YAKZ	Yakutsk, Russia	105.3	TRIAL	−53.2	4.3	2.4		5	
YAR1	Mingenew, Australia	267.1	TRIAL	27.4	—	—			
YAR2	Dongara, Australia	267.1	TRIAL	27.4	—	—			
YELL	Yellowknife, Northwest Territories, Canada	207.9	SYNO	27.4	—	—			P
YSSK	Yuzhno-Sakhalinsk, Russia	66.1	TRIAL	−128.7	42.1	12.2			P
ZIMM	Zimmerwald, Switzerland	907.9	GPS	−161.0	416.9	39.7			
ZWEN	Zwenigorod, Russia	254.6	TRIAL	77.4	54.6	74.7			P

(GPS site height). In addition, incorrectly reported GPS site elevations will contribute to bias differences.

Figure 1 illustrates the GPS_PW distribution of GSD and IGS observations. The GSD GPS_PW observations range between 0 and 55 mm. Most of the observations (98.3%) are less than 35 mm (only 158 observations have PW > 35 mm). This reflects the relatively

cold (hence, low moisture) Canadian climate. GSD PW observations greater than 35 mm are mostly from the month of July, when warmer, more humid air masses are found over regions of the country. For the IGS GPS_PW observations, 86.0% of the observations are in the 0–35-mm range, but there are still a large number of observations (≈ 8080) that are greater than 35 mm.

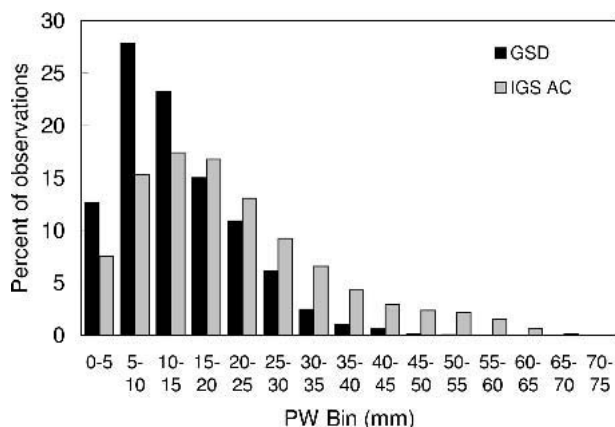


FIG. 1. Distribution of GPS_PW (mm) observations for both the GSD and IGS AC datasets. For the GSD dataset, there are 9078 observations (20 GPS sites and 4 months: Apr, Jul, and Oct 2001, and Jan 2002). For the IGS AC dataset, there are 57 518 observations (112 GPS sites and 7 months: Apr, Jun, Jul, Aug, Sep, and Oct 2001, and Jan 2002). Only observations at ANAL/TRIAL times are considered (i.e., 0000, 0600, 1200, and 1800 UTC).

a. GPS_PW compared with RS_PW

1) GSD GPS SITES

GPS_PW is compared with RS_PW at eight sites (Table 3). Churchill is the most ideal collocation, with the smallest differences in site location and height. Overall, there is close agreement between GPS_PW and RS_PW. The correlation coefficient, SD, and bias (GPS_PW minus RS_PW) are, respectively, 0.97, 2.04 mm, and 1.35 mm (Table 5). These values are in agreement with those available in the literature (listed in section 1) and, in particular, with those obtained by Yang et al. (1999) for RS stations located in Sweden and Finland where weather conditions are similar to those in Canada. Table 6 lists the bias and SD for each site for January 2002 (winter) and July 2001 (summer). The bias and SD are both higher for the summer month, with a higher mean PW.

The dependence of the bias and SD on PW for PW less than 30 mm is shown in Fig. 2. Both tend to increase with PW. The tendency for the SD to increase

with PW was also found in Haase et al. (2003), Basili et al. (2001), and Wolfe and Gutman (2000), whereas the tendency of the bias to increase with PW was observed in Ohtani and Naito (2000). The dependence of SD on PW could be attributed, in part, to stronger humidity gradients that can exist between dry and moist air when moister air is involved (Basili et al. 2001; Ohtani and Naito 2000). In the presence of strong gradients, the location and sampling differences between GPS and RS can be more significant than for lower PW conditions. In addition, the presence of strong horizontal gradients in atmospheric properties can have a negative impact on ZTD accuracy (i.e., Ohtani and Naito 2000) due to a breakdown of the azimuthal symmetry assumption. Moreover, a given error in radiosonde RH measurement will lead to a linear increase in absolute PW error with PW (because $RH = q/q_s$, where q_s is the saturation specific humidity and only depends on pressure and temperature).

Radiosonde measurement errors are described in detail in Wang et al. (2002). In Canada, all of the radiosondes launched after 1998 are the Vaisala RS80-H (T. Eperson 2003, personal communication). The radiosonde manufacturer Vaisala has implemented corrective measures to eliminate a known dry bias in the humidity measurements from their RS80-H radiosonde that is caused by chemical contamination. Prior to implementing the corrective steps, the dry bias could reach 10% at saturation for 1-yr-old (time after the date of manufacture) radiosondes. At most Canadian radiosonde sites, sondes with the applied corrective measures were put into use in January 2002. This coincides with the time of the last month of data obtained from GSD. Therefore, there is a known dry bias for a large part of the RS data over Canada, but quantifying this bias (Wang et al. 2002) is beyond the scope of this paper. This RS dry bias could partly explain the moist bias of GPS_PW with respect to RS_PW found in this study.

2) IGS GPS SITES

GPS_PW is compared with RS_PW at 66 GPS sites (Table 4). There are 34 sites with GPS Met P_s (11) or SYNO P_s (23) observations, and 32 sites with only

TABLE 5. Intercomparison statistics (r = correlation, SD = standard deviation, Bias = difference of means, N = No. of observations) for GPS_PW vs RS_PW, ANAL_PW, and TRIAL_PW for the GSD and IGS datasets. PW units are in millimeters.

GPS ZTD dataset	Data compared	r	SD	Bias	Mean GPS_PW	N
GSD*	GPS – RS**	0.97	2.04	1.35	14.9	1664
GSD	GPS – ANAL	0.97	2.03	1.19	13.5	9078
GSD	GPS – TRIAL	0.96	2.22	1.02	13.5	9078
IGS*	GPS – RS**	0.98	2.60	0.67	20.8	19 967
IGS	GPS – ANAL	0.97	3.45	0.65	21.1	57 518
IGS	GPS – TRIAL	0.96	3.55	0.52	21.1	57 518

* The statistics for the GSD data are over the 4 months of data, and those for the IGS dataset are over the 7 months of data.

** GPS_PW and RS_PW comparisons are at 0000 and 1200 UTC. GPS_PW and ANAL_PW or TRIAL_PW comparisons are at 0000, 0600, 1200, and 1800 UTC.

TABLE 6. GSD GPS_PW minus RS_PW comparison statistics by site for Jan 2002 and Jul 2001. Units of PW are in millimeters.

Site	Jan 2002			Jul 2001		
	Mean GPS_PW	Bias	SD	Mean GPS_PW	Bias	SD
CHUR	4.3	1.23	0.40	22.8	2.02	1.21
NRC1	8.1	0.65	0.95	24.9	1.67	3.30
STJO	7.7	1.18	0.85	25.2	1.65	1.64
ALBH	12.1	-0.27	1.76	21.7	2.30	3.66
CAGS	8.3	1.24	0.87	24.5	2.75	2.44
ELIZ	12.9	1.22	1.85	22.2	1.57	2.53
BCOV	13.0	0.66	1.10	24.0	2.33	2.07
HOLB	Missing data			18.1	0.63	2.10

TRIAL P_s . As was found for the GSD sites, there is very good overall agreement between GPS_PW and RS_PW. For all 66 sites, the correlation coefficient, SD, and bias (GPS_PW minus RS_PW) are, respectively, 0.98, 2.60 mm, and 0.67 mm (Table 5). The SD for the IGS dataset is higher than that of the GSD dataset, but the mean PW of the former is also higher by 5.9 mm. As was found for the GSD sites, there is an overall moist bias of GPS_PW compared to RS_PW. However, unlike the GSD GPS sites results, for which only two sites had a small negative bias (for one given month each), several GPS sites have a near-zero or negative (dry) 7-month bias. This might, in part, be related to the different types of sondes used by different countries, because sondes from different manufacturers have unique moisture measurement errors and biases [e.g., section 7a(1)].

The variation of the bias and SD with PW is shown in Fig. 3. Figure 3a shows the bias and SD for the 34 sites with GPS Met/SYNO P_s (section 6), while the results for the 32 sites with TRIAL P_s are presented in Fig. 3b. The overall moist bias of GPS_PW relative to RS_PW is evident over the entire range of PW for both sets of sites. There is a slight increase in bias with the PW for

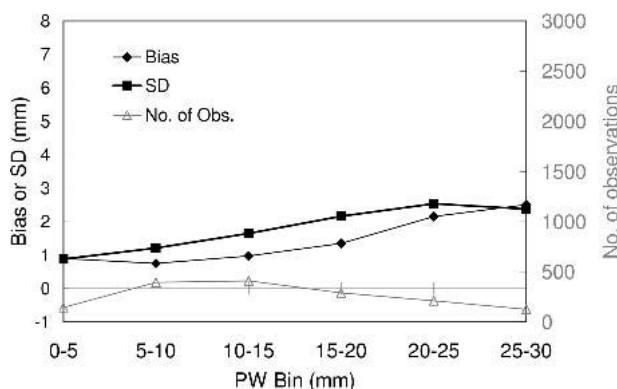


FIG. 2. GSD GPS_PW minus RS_PW bias, SD, and number of data collocations in each GPS_PW bin (8 GPS sites and 4 months: Apr, Jul, and Oct 2001 and Jan 2002). Comparisons are done at RS observation times of 0000 and 1200 UTC.

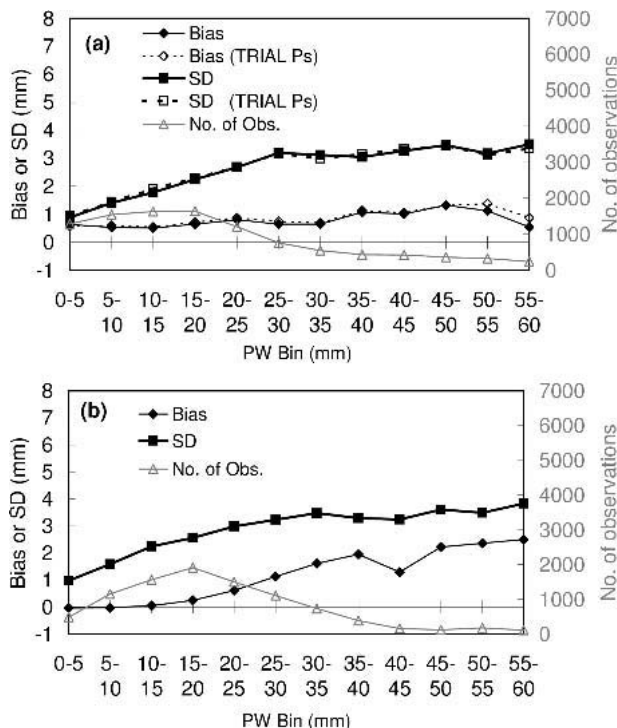


FIG. 3. IGS GPS_PW minus RS_PW bias, SD, and number of data collocations in each GPS_PW bin (66 GPS sites with collocated RS for all 7 months). Comparisons are done at the RS observation times of 0000 and 1200 UTC: (a) 34 GPS sites with GPS Met or SYNO P_s ; (b) 32 GPS sites with TRIAL P_s . Curves labeled Bias(TRIAL P_s) and SD(TRIAL P_s) in (a) are results when GPS_PW is computed from TRIAL P_s rather than GPS Met/SYNO P_s .

the set with GPS Met/SYNO P_s (Fig. 3a)—from ≈ 0.6 mm at low PW to ≈ 1.2 mm at higher PW. There is a much more pronounced increase in bias for the set with TRIAL P_s (Fig. 3b)—from near zero at low PW to ≈ 2.6 mm at high PW. For both sets of sites, the SD increases from ≈ 1 to ≈ 4 mm over the PW range (0–60 mm), with slightly higher SD values in Fig. 3b (GPS_PW computed with TRIAL P_s). The tendency for the PW bias and SD to increase with PW is also observed, as is the case for the GSD data.

To examine the effects of using either TRIAL or GPS Met/SYNO P_s as a surface pressure observable, TRIAL P_s is substituted for GPS Met/SYNO P_s in re-computing GPS_PW for the sites in Fig. 3a. The comparison results for this case are also shown in Fig. 3a. There is little change in the bias and SD, which suggests that TRIAL P_s and GPS Met/SYNO P_s are in close agreement for this set of sites when averaged over the 7-month period. This is not unexpected, because SYNO observations available at 8 of the 11 GPS Met sites are assimilated into the analyses that are used as initial conditions for the TRIAL forecast. However, no P_s observations are available for assimilation at the GPS site locations of Fig. 3b. TRIAL P_s accuracy at these

sites is expected to be lower than at the sites of Fig. 3a and could contribute to the differences noted above between Figs. 3a and 3b, namely, a higher bias for PW above 30 mm at the sites in Fig. 3b, with a slightly higher SD for all ranges of PW.

b. GPS_PW compared with ANAL_PW and TRIAL_PW

1) GSD GPS SITES

GPS_PW is compared with TRIAL_PW and ANAL_PW for all 20 GPS sites. Comparison with ANAL_PW gives a correlation coefficient, SD, and bias (GPS_PW minus ANAL_PW) of 0.97, 2.03 mm, and 1.19 mm, respectively (Table 5). These statistics are close to those of the GPS_PW versus RS_PW intercomparison (i.e., 0.97, 2.04 mm, and 1.35 mm). It should be noted, however, that the mean GPS_PW is 1.4-mm lower in the GPS_PW versus ANAL_PW collocation dataset than that of the GPS_PW versus RS_PW collocation dataset (Table 5). Seventeen of the 20 GPS sites (i.e., 85%) have a RS station located within 300 km, which is the typical radius of influence for assimilated RS data (Laroche et al. 1999). This largely explains why the above statistics are close.

Comparison statistics of GPS_PW with TRIAL_PW (Table 5) (i.e., correlation, SD, and bias, respectively) are 0.96, 2.22 mm, and 1.02 mm. The SD of GPS_PW versus TRIAL_PW is 0.19-mm higher than that of GPS_PW versus ANAL_PW. This reflects the presence of forecast errors in the TRIAL field that are later reduced through the assimilation of RS data in producing the ANAL field. The dependence of bias and SD on PW is shown in Fig. 4. The lower bias of GPS_PW minus TRIAL_PW may be due to the fact that TRIAL_PW is based on forecast data only, which, unlike the analysis, has not been altered through assimilation of RS data with a possible dry bias [section 7a(1)]. As with GPS_PW versus RS_PW (section

7a), there is an overall increase with PW in both bias and SD for GPS_PW versus both ANAL_PW and TRIAL_PW. Possible reasons for these are the same as those suggested in section 7a(1).

2) IGS GPS SITES

GPS_PW versus TRIAL_PW and ANAL_PW statistics are computed for 82 GPS sites. There are 33 sites with GPS Met/SYNO P_s (13 GPS Met, 20 SYNO) and 49 sites with only TRIAL P_s . The correlation between the GPS_PW and TRIAL_PW is 0.96, while the SD is 3.55 mm. The errors associated with 6-h forecast (TRIAL) moisture fields account for the lower correlation and higher SD (by about 1 mm) compared to those computed for GPS_PW versus RS_PW (i.e., 0.98 and 2.60 mm, respectively). Occasional large differences (>15 mm or 4σ) are observed between GPS_PW and TRIAL_PW, in contrast to the GPS_PW versus RS_PW intercomparison. Such cases are mainly restricted to seven tropical and subtropical GPS sites [i.e., Miami, Florida (AOML), Bermuda (BRMU), Cocos, Australia (COCO), Dededo, Guam (GUAM), Kwajalein Atoll, Marshall Islands (KWJ1), Papeete, Tahiti (THTI), and Eusebio, Brazil (FORT)]. At three of these sites, differences as high as 30 mm occur. GPS_PW minus TRIAL_PW SD for each of the seven sites is in the 5–6-mm range, higher than the overall SD of 3.55 mm. Examination of sites with large differences reveals the presence of dry layers in the middle and upper troposphere, which are not well represented in the TRIAL fields. Such features, typically formed by subsidence, can create variability in PW comparable to that observed for airmass changes in the midlatitudes.

The GPS_PW minus ANAL_PW SD is 3.45 mm (Table 5), which is only 0.11 mm lower than the GPS_PW minus TRIAL_PW SD (3.55 mm). Sixty of the 82 GPS sites (i.e., 73%) have a RS station located within 300 km. This ratio is quite large. However, RS observations that are significantly different from the 6-h forecast (TRIAL) are likely to be rejected in areas where forecasts errors are high (i.e., Tropics and subtropics). The rejection process is part of the QC of the radiosonde data (appendix B). In addition, some RS stations report data with low and/or variable quality.

The variation of bias and SD with PW for GPS_PW minus TRIAL_PW and GPS_PW minus ANAL_PW are shown in Fig. 5a (for sites with GPS Met/SYNO P_s) and Fig. 5b (for sites with TRIAL P_s). The SD (which increases with PW) is considerably higher for the GPS_PW versus TRIAL_PW and GPS_PW versus ANAL_PW than for the GPS_PW versus RS_PW (Fig. 3), especially for PW greater than 25 mm, for reasons discussed above. The SD values are slightly lower for GPS_PW versus ANAL_PW than for GPS_PW versus TRIAL_PW. This difference is most evident for PW between 25 and 55 mm for sites with GPS Met/SYNO P_s (Fig. 5a).

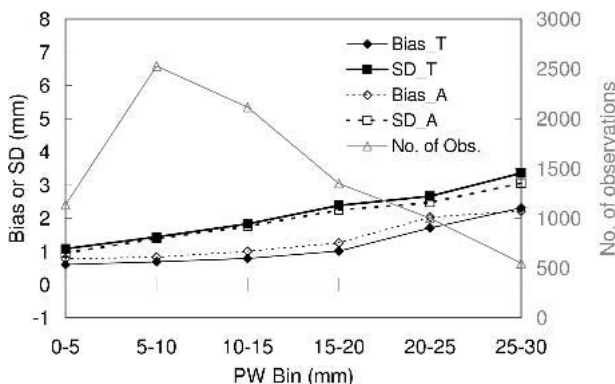


FIG. 4. As in Fig. 2, but for GSD GPS_PW minus ANAL_PW (A) and GPS_PW minus TRIAL_PW (T). Statistics are shown for all 20 sites at times of 0000, 0600, 1200, and 1800 UTC.

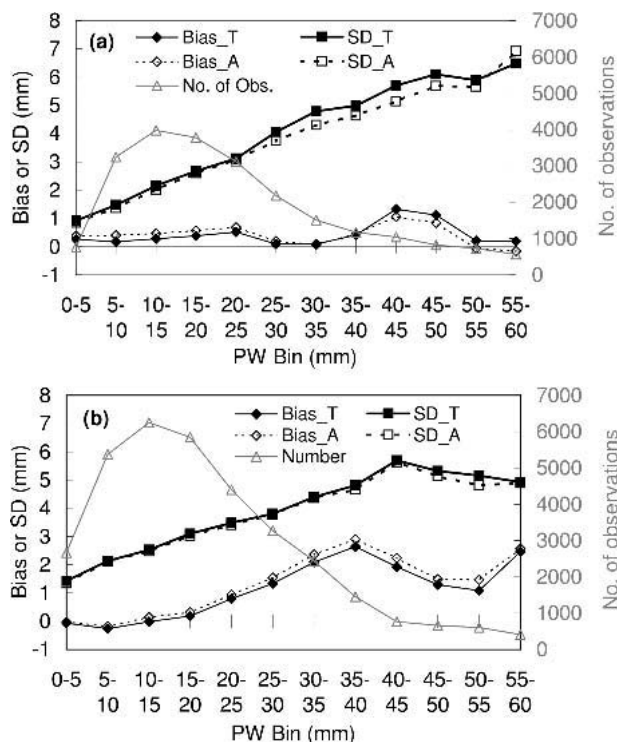


FIG. 5. As in Fig. 3, but for GPS_PW minus TRIAL_PW (T) and GPS_PW minus ANAL_PW (A). Observations are at the TRIAL/ANAL times of 0000, 0600, 1200, and 1800 UTC: (a) 33 GPS sites with GPS Met or SYNO P_s ; (b) 49 GPS sites with TRIAL P_s .

c. Estimates of GPS_PW, RS_PW, and TRIAL_PW errors

By comparing PW from three independent sources (GPS, RS, and TRIAL) (referred to as a three-way intercomparison), it is possible to compute the PW error that is associated with each data source (appendix C). The observation sample must be large enough to be statistically significant. In addition, there can be no correlation between the errors of each source of data. The intercomparison is done at 19 IGS sites (Table 4) where all three sources of PW are available. To ensure that GPS_PW error is uncorrelated with that of TRIAL_PW, GPS_PW is computed with GPS Met P_s (8 sites) or SYNO P_s (11 sites). Furthermore, T_m is computed from GPS Met T_s (rather than using TRIAL T_m) with the Bevis et al. (1992) regression equation. The PW errors are estimated based on 5655 data collocations at the 19 IGS sites and are only made for PW observations in the 0–55-mm range due to the relatively low number of observations with PW greater than 55 mm. The 7-month-averaged GPS_PW for the 19 IGS sites is 26 mm, and the estimated GPS_PW, RS_PW, and TRIAL_PW errors are, respectively, 1.97, 1.82, and 3.69 mm. Overall PW error is smallest for RS_PW and greatest for TRIAL_PW with GPS_PW error very close to that of the RS_PW error.

The dependence of PW error on PW is studied by grouping the PW data into bins and performing the error estimation separately for each bin. The results are shown in Fig. 6 in terms of absolute and relative error. Absolute GPS_PW error is relatively constant at ≈ 1 mm up to the 20–25-mm bin (Fig. 6a), whereas RS_PW and TRIAL_PW error increase in a linear fashion from ≈ 1 to ≈ 2 mm for the former and ≈ 1 to ≈ 3 mm for the latter. Beyond the 20–25-mm bin, the GPS_PW error increases to 2–3 mm. The RS_PW error stays constant around 2 mm, while the TRIAL_PW error reaches 5.4 mm for the 50–55-mm bin. The lower accuracy of TRIAL_PW when compared with GPS_PW and RS_PW is especially pronounced for PW greater than 25 mm. Some of these higher-PW observations come from GPS sites located in warm and humid tropical and subtropical regions of the world. For reasons discussed above [section 7b(2)], accuracy of the TRIAL moisture fields is generally lower for such regions. Relative PW error for all three data sources (Fig. 6b) is highest for very low PW (0–5 mm) and is 16% for RS_PW, 28% for TRIAL_PW, and 33% for GPS_PW. Thus, for dry atmospheres in particular, it is important to have accurate measurements of surface pressure collocated with the GPS antenna (appendix A) to minimize the relative error of GPS_PW.

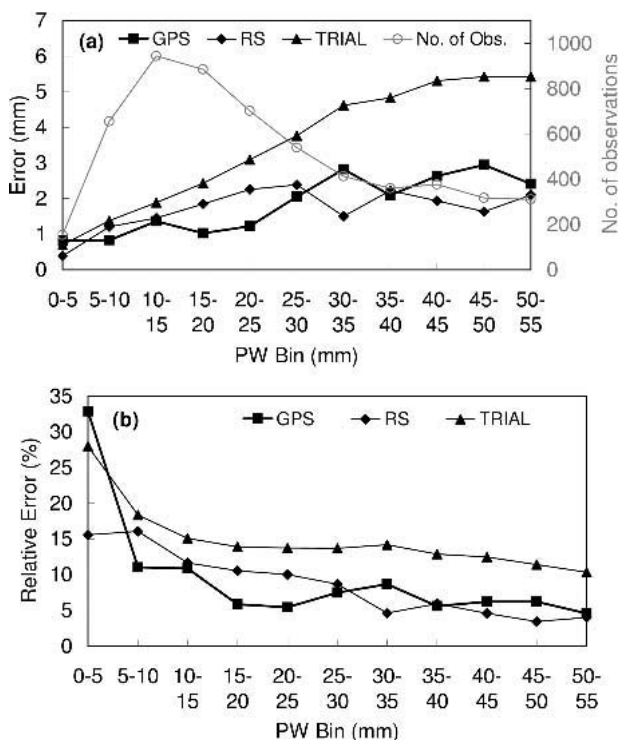


FIG. 6. Estimated error of GPS_PW, RS_PW, and TRIAL_PW as a function of PW (19 IGS sites and 7 months of data at 0000 and 1200 UTC) based on a three-way intercomparison (section 7c): (a) absolute error and number of observations; (b) relative error.

8. Conclusions

The objective of this study is to evaluate PW derived from GPS ZTD (computed with the highest accuracy final orbits) coming from two different datasets. The first ZTD dataset was provided by GSD/NRCan for 20 GPS sites located in Canada (April, July, and October 2001, and January 2002), and for the second dataset, data were obtained for 112 GPS sites from the IGS global tracking network distributed over the globe (April, June, July, August, September, and October 2001, and January 2002). These two datasets were studied to cover a wide range of PW values (i.e., from arctic or very dry to tropical or very humid profiles). The evaluation study of GPS_PW is the first step toward eventual DA of ZTD in a 4DVAR system (to take full advantage of their high temporal resolution) and is done through the intercomparison of PW computed from radiosondes, operational analyses, and GEM model 6-h forecasts produced by the CMC global 3DVAR system.

Ninety-eight percent of GSD GPS_PW observations are in the 0–35-mm range (reflecting the relatively cold Canadian climate), whereas 86% of the IGS GPS_PW (with globally distributed sites) are in the same range. For the GSD data (mean GPS_PW of 14.9 mm), there is very good agreement (correlation of 0.97) between GPS_PW and RS_PW with an overall moist bias (GPS_PW minus RS_PW) of 1.35 mm and a SD of 2.04 mm. For the IGS data (mean GPS_PW of 20.8 mm), the correlation, bias, and SD are, respectively, 0.98, 0.67 mm, and 2.6 mm. Bias and SD are found to increase with PW. The correlation, bias, and SD results are within the range obtained by previous published studies. For the GSD data, the comparison statistics of GPS_PW with ANAL_PW are very similar to those of the GPS_PW versus RS_PW comparison. This is not surprising because the ANAL_PW is heavily based on assimilated RS data. The degree of agreement is less for the IGS dataset because the analyses are less influenced by the RS data (due to a sparser network of RS observations available for assimilation and a higher rejection of RS data available for assimilation due to poorer 6-h forecasts in the Tropics and subtropics). As expected, comparisons of GPS_PW with 6-h-forecast PW (TRIAL_PW) instead of ANAL_PW (at both the GSD and IGS sites) show a lower correlation and higher SD because the analyses are more accurate than the TRIAL fields.

From a three-way intercomparison of GPS_PW, RS_PW, and TRIAL_PW at 19 IGS sites over 7 months, it is found that GPS_PW has the lowest estimated error (≈ 1 mm) for PW between 5 and 30 mm. For PW greater than 30 mm, the RS_PW estimated error is ≈ 2 mm, and that of GPS_PW is slightly higher at ≈ 2.5 mm. The TRIAL_PW estimated error is the largest of the three—except for very low PW (0–5

mm)—and increases with PW to reach 5.4 mm in the 50–55-mm PW range (or 10% in relative error).

The relative estimated error of GPS_PW for very dry conditions (PW of 0–5 mm) is high at 33%. To minimize the GPS_PW retrieval error in dry/cold air masses (similar to those that are frequently observed over Canada), accurate surface pressure observations (GPS Met or SYNO) should be available at the GPS sites. Surface pressure error from the TRIAL (6-h forecast) can be much larger than that of GPS Met or SYNO, especially for sites in data-sparse areas where there are few surface observations to assimilate and/or where forecast P_s is poor due to topography or diurnal effects.

It is expected that the CMC regional 3DVAR system (covering North America and the surrounding waters) will benefit from the several hundred GPS ZTD observations that are available over North America, while the high temporal resolution of ZTD make it ideal for use with the 4DVAR system that will be implemented at CMC by the end of the year 2004. A monitoring system (comparing ZTD observations and ZTD computed from GEM model 6-h forecasts and analyses) for the near-real-time ZTD product (e.g., Ge et al. 2002; Douša 2001; Gendt et al. 2004) that is available from NOAA/FSL (Gutman et al. 2003a) is currently being developed at the Meteorological Service of Canada. This activity will be followed by actual data assimilation experiments.

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APPENDIX A

Quality Control and Instrument Accuracy of Surface Pressure Observations

The QC of GPS Met and SYNO P_s observations is done by comparison with the GEM model TRIAL P_s field. All P_s values are adjusted hydrostatically to the GPS antenna elevation. To remove gross errors, P_s observations are rejected when 1) the range is outside 600 and 1080 hPa, and 2) when the absolute difference with TRIAL P_s is greater than 30 hPa.

The barometer (Vaisala PTB100) at the Canadian GPS sites measures surface pressure with an accuracy of 0.25 hPa (M. Bujold 2003, personal communication). However, if the barometer is at a different height and/or location than that of the GPS antenna, P_s accuracy is estimated to be less (e.g., <0.3 hPa according to Baltink et al. 2002). At the Canadian synoptic weather stations, the instrument accuracy for pressure measurements at automatic stations is ± 0.3 hPa (Setra 270 pressure transducer) and is ± 0.15 hPa at manned stations (Vaisala PTB220 class A barometer) (B. Sheppard and K. Devine 2003, personal communication).

For the GSD data (over the 4 months), SD of GPS

Met P_s minus ANAL P_s for each of the nine Canadian sites (the Whitehorse site had problematic data) is between 0.22 and 0.45 hPa. The SD of GPS Met P_s minus TRIAL P_s is between 0.66 and 1.40 hPa. As expected, ANAL P_s is more accurate than TRIAL P_s —the latter being a 6-h forecast. SYNO P_s is available at 12 of the GSD GPS sites, including 8 of the 9 sites that have GPS Met P_s . The SD of GPS Met P_s minus SYNO P_s is between 0.13 and 0.44 hPa, showing a high level of agreement.

APPENDIX B

Radiosonde Data Quality Control

Data from RS ascents are reported at so-called mandatory and significant pressure levels. Mandatory levels must always be reported. The actual raw RS data are transmitted by the sonde during its ascent with a significantly higher vertical resolution but are not transmitted over the global telecommunication system (GTS) because of earlier bandwidth limitations. As a minimum requirement for RS_PW to be computed, RS data, after having passed QC, must be available at all mandatory levels for pressure greater or equal to 250 hPa (i.e., surface, 1000 hPa, 925 hPa, 850 hPa, 700 hPa, 500 hPa, 400 hPa, 300 hPa, and 250 hPa).

Initial QC of RS data performed at CMC is based on simple checks against climatological extremes and pressure/temperature consistency checks. For this study, additional QC is performed by removing poor data levels with a background check (comparison of observations at each RS level against GEM model TRIAL fields). The quality of RS data from North America is such that few entire soundings are rejected by the additional QC. Many more soundings are rejected over the rest of the world (except for Europe).

APPENDIX C

Estimation of Errors for GPS, RS, and TRIAL Precipitable Water

Each measure of precipitable water (PW) has an associated error:

$$PW_{GPS} - \overline{PW_{GPS}} = PW_t - \overline{PW_t} + E_{GPS}, \quad (C1)$$

$$PW_{RS} - \overline{PW_{RS}} = PW_t - \overline{PW_t} + E_{RS}, \quad \text{and} \quad (C2)$$

$$PW_{TRIAL} - \overline{PW_{TRIAL}} = PW_t - \overline{PW_t} + E_{TRIAL}, \quad (C3)$$

where PW_t is the true value of PW, the overbar represents the ensemble mean, and the E terms are the errors (with zero mean) associated with each measurement. Taking differences between paired equations above gives

$$PW_{GPS} - \overline{PW_{GPS}} - (PW_{TRIAL} - \overline{PW_{TRIAL}}) = E_{GPS} - E_{TRIAL}, \quad (C4)$$

$$PW_{GPS} - \overline{PW_{GPS}} - (PW_{RS} - \overline{PW_{RS}}) = E_{GPS} - E_{RS}, \quad (C5)$$

and

$$PW_{TRIAL} - \overline{PW_{TRIAL}} - (PW_{RS} - \overline{PW_{RS}}) = E_{TRIAL} - E_{RS}. \quad (C6)$$

Squaring Eqs. (C4)–(C6), computing their mean, and assuming that cross terms of error are uncorrelated, gives

$$SD_{GPS-TRIAL}^2 = \overline{E_{GPS}^2} + \overline{E_{TRIAL}^2}, \quad (C7)$$

$$SD_{GPS-RS}^2 = \overline{E_{GPS}^2} + \overline{E_{RS}^2}, \quad \text{and} \quad (C8)$$

$$SD_{TRIAL-RS}^2 = \overline{E_{TRIAL}^2} + \overline{E_{RS}^2}. \quad (C9)$$

One can combine the three equations above and solve for $(\overline{E_{RS}^2})^{1/2}$, $(\overline{E_{TRIAL}^2})^{1/2}$, and $(\overline{E_{GPS}^2})^{1/2}$, which are, respectively, estimates of the RS_PW, TRIAL_PW, and GPS_PW error.

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